THRESHOLD DISCREPANCY BARRIERS FOR DEPTH-THREE SCALAR PLANTED-CLIQUE AMP SCHEDULES

ARGUS OPEN RESEARCH PROJECT

Abstract. For the planted-clique recovery problem at clique size $k = \lceil \sqrt{n}/\log n \rceil$, we prove two exact limitations on frozen scalar approximate message passing (AMP) / cavity transfer schedules: (1) uniform positive-semidefinite domination via the collision Gram matrix alone fails at reachable parity states; and (2) every admissible rooted-C_4, depth-three scalar score has root-in/root-out threshold discrepancy $O(1/\log n + n^{-1/2})$, preventing sub-$\sqrt{n}$ candidate isolation.

1. Main Barriers

Theorem 1.1. *At $k = \lceil \sqrt{n}/\log n \rceil$ in $G(n, 1/2, k)$:*

(1) *Under admissible Lipschitz scalar denoisers, a reachable parity configuration sends a zero root-out shifted-Gram block to a nonzero rank-one all-ones block, ruling out uniform PSD pullback domination by the collision Gram matrix.*

(2) *For the rooted-C_4 depth-three scalar message schedule, signed diagonal cancellation incurs a loss of $k - 1$ and bounds the root-in versus root-out score discrepancy by $O(1/\log n + n^{-1/2}) = o(1)$.*

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